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Turbulence as an Unidentifiability Theorem: 60|Indistinguishability of N-Body Truncation Error and True Turbulence at Finite Resolution 61|Turbulence as an Unidentifiability Theorem: 62|Indistinguishability of Truncation Error and True Turbulence 63|under Finite-Resolution N-Body Dynamics

SCX Theory Group

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Abstract

72|We propose a fundamental epistemological reconstruction: turbulence is not a closure problem, but an **unidentifiability problem** 73| —
 $\mathbf{F} = m\mathbf{a}$ N MD $N \rightarrow \infty$ Boltzmann Navier–Stokes NS
10²³ 74| 75|This paper presents three theorems. 1
RANS — $\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq C \cdot h^\alpha$ $h \rightarrow 0$ $\text{TV} \rightarrow 0$ $h > 0$ $\text{TV} >$
0 2 M RANS $k\text{-}\varepsilon$, $k\text{-}\omega$ SST, LES, DES, DNS Δ systematic bias
 $\mathbb{P}(\cap_{m=1}^M \{B_m > \Delta\}) \leq \exp(-2M\Delta^2/L^2)$ DNS 3 Kolmogorov
 h $E_h(k) = E_{\text{true}}(k) + C_{\text{res}} \cdot h^{2/3} \cdot k^{-5/3} + C_{\text{stat}} \cdot N^{-1/2} + o(h^{2/3} + N^{-1/2})$ —5/3
—Kolmogorov 76| 77|The corollary: turbulence modeling is not about
better closure models — it is about declaring your **resolution assumption**. At different
resolutions, turbulence is different things. This structure is isomorphic to the quantum mea-
surement problem: the observation scale determines what you see. 78|

79| 80|81|

1 Starting Point: A Forgotten Fact

82|83| 84|Every fluid dynamicist learns the Navier–Stokes equations: 85|

$$86| \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} 87| = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}, 88| \quad (1)$$

89|and that these emerge from Boltzmann via Chapman–Enskog. Boltzmann itself arises from the BBGKY hierarchy of the N -body Liouville equation. The chain bottom is Newton: 90|

$$91| m_i \frac{d^2 \mathbf{x}_i}{dt^2} = \mathbf{F}_i 92| = - \sum_{j \neq i} \nabla_{\mathbf{x}_i} V(|\mathbf{x}_i - \mathbf{x}_j|), 93| \quad (2)$$

94| $V(r)$ Lennard-Jones $i = 1, \dots, N$ $N \sim 10^{23}$ 95| 96|Key: Equation (??) is exact within classical mechanics. If you solved 10²³ coupled ODEs exactly, you would obtain **the complete flow field at all scales** — including all fluctuations we call “turbulence.” 97| 98|But you cannot. 99| 100|This is the root. Physics is not missing anything — it is **the fundamental limit of computation** preventing a first-principles solution. This is not engineering — it is **epistemological** 101| 102|103|

2 From N-Body MD to Continuum: The Convergence Chain

104|105| 106|

2.1 Micro $\rightarrow$ Meso $\rightarrow$ Macro

107| 108|Define the microscopic phase-space density: 109|

$$110|f_N(\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{p}_1, \dots, \mathbf{p}_N, t), 111| \quad (3)$$

112|satisfying $\partial_t f_N = \{H, f_N\}$. Integrating over $N - k$ coordinates: 113|

$$114|f^{(k)}(\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{p}_1, \dots, \mathbf{p}_k, t) 115| = \int f_N \, d\mathbf{x}_{k+1} \dots d\mathbf{x}_N \, d\mathbf{p}_{k+1} \dots d\mathbf{p}_N. 116| \quad (4)$$

117| 118|This yields the BBGKY hierarchy. In the Boltzmann–Grad limit ($N \rightarrow \infty$, $d \rightarrow 0$, $Nd^2 = \text{const}$, $d = \text{effective diameter}$), $f^{(1)}$ satisfies Boltzmann. Chapman–Enskog Euler Navier–Stokes 119| 120|**Convergence Proposition:** 121|

$$122| \lim_{N \rightarrow \infty} \mathbf{u}_N(t, \mathbf{x}) = \mathbf{u}_{\text{NS}}(t, \mathbf{x}), 123| \quad (5)$$

124|where $\mathbf{u}_N$ is the coarse-grained velocity from N -body MD (averaged over Δx , Δt), $\mathbf{u}_{\text{NS}}$ is a NS solution. 125| 126|

2.2 The Coarse-Graining Operator

127| 128| The Coarse-Graining Operator $\mathcal{C}_h$ 129|

$$130|(\mathcal{C}_h \mathbf{v})(t, \mathbf{x}) 131| = \frac{1}{|B_h(\mathbf{x})|} 132| \int_{B_h(\mathbf{x})} \frac{1}{N_h(\mathbf{y})} \sum_{i: \mathbf{x}_i(t) \in B_h(\mathbf{y})} \dot{\mathbf{x}}_i(t) 134| \, d\mathbf{y}, 135| \quad (6)$$

136| $B_h(\mathbf{x})$ $\mathbf{x}$ h $N_h(\mathbf{y})$ $B_h(\mathbf{y})$ 137|Then: 138|

$$139|\mathbf{u}_N(t, \mathbf{x}) = (\mathcal{C}_h \dot{\mathbf{X}})(t, \mathbf{x}), 140| \quad (7)$$

141|where $\dot{\mathbf{X}} = (\dot{\mathbf{x}}_1, \dots, \dot{\mathbf{x}}_N)$ is the N -body phase-space trajectory. 142| 143|144|

3 Error Decomposition and Scaling Analysis at Finite Resolution

145|146| 147|This section promotes scaling analysis from appendix to main text. 148| 149|

3.1 Definition of Three Velocity Fields

150| 151|We define three **conceptually distinct** velocity fields: 152| 153|

154|

(1) $\mathbf{u}_N(t, \mathbf{x})$ — N MD N h 155|

(2) $\mathbf{u}_{\text{NS}}(t, \mathbf{x})$ — Navier–Stokes 156|

(3) $\mathbf{u}_{\text{RANS}}(t, \mathbf{x})$ — RANS 157|

158| 159|Note: $\mathbf{u}_{\text{NS}}$ **already contains turbulence** — if NS indeed describes turbulence. $\mathbf{u}_{\text{RANS}}$ is the **smoothed field** after closure — its turbulence has been modeled away. 160| 161|

3.2 Error Decomposition

162| 163|For finite N and finite h : 164|

$$165| \|\mathbf{u}_N - \mathbf{u}_{\text{NS}}\|_{L^2} 166| = \|(\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]) + (\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}})\|_{L^2} 167| \leq \|\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]\|_{L^2} + \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2}. 168| \quad (8)$$

169| 170|Define: 171|

$$172|\varepsilon_{\text{stat}} := \|\mathbf{u}_N - \mathbb{E}[\mathbf{u}_N]\|_{L^2} 173| = \quad \text{statistical fluctuation} , \quad (9)$$

$$174|\varepsilon_{\text{res}} := \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2} 175| = \quad \text{resolution truncation error} . 176| \quad (10)$$

177| 178|

3.3 Spectral Representation and Scaling of Truncation Error

179| 180| $\mathbf{u}_{\text{NS}}$ Fourier $\hat{\mathbf{u}}(k)$ The Coarse-Graining Operator $\mathcal{C}_h$ $\hat{G}_h(k)$ 181|

$$182|\widehat{\mathcal{C}_h \mathbf{u}_{\text{NS}}}(k) = \hat{G}_h(k) \hat{\mathbf{u}}(k), 183| \quad (11)$$

184| $\hat{G}_h(0) = 1$ $\hat{G}_h(k) \rightarrow 0$ $k \gg 2\pi/h$ Gaussian $\hat{G}_h(k) = e^{-(kh)^2/2}$ $\hat{G}_h(k) = \text{sinc}(kh/2)$ 185| 186|The L^2 norm of truncation error is, by Parseval: 187|

$$188|\varepsilon_{\text{res}}^2 = \|\mathbb{E}[\mathbf{u}_N] - \mathbf{u}_{\text{NS}}\|_{L^2}^2 189| = \int_{\mathbb{R}^3} \left| 1 - \hat{G}_h(k) \right|^2 E(k) \, dk, 190| \quad (12)$$

191|where $E(k)$ is the energy spectrum. For Kolmogorov $E(k) = C_K \epsilon^{2/3} k^{-5/3}$, Gaussian filter, $x = kh$: 192|

$$193|\varepsilon_{\text{res}}^2 \sim C_K \epsilon^{2/3} \int_0^\infty (1 - e^{-(kh)^2/2})^2 k^{-5/3} \, dk 194| = C_K \epsilon^{2/3} h^{2/3} \int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx. 195| \quad (13)$$

196|Since $\int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx < \infty$: 197|

$$198|\varepsilon_{\text{res}} \sim C_{\text{res}} \cdot h^{1/3}, 199| \quad C_{\text{res}} = \sqrt{C_K} \epsilon^{1/3} \left(\int_0^\infty (1 - e^{-x^2/2})^2 x^{-5/3} \, dx \right)^{1/2}. 200| \quad (14)$$

201|Generally, if $E(k) \sim k^{-\beta}$, then $\varepsilon_{\text{res}} \sim h^{(\beta-1)/2}$. 202| 203|

3.4 Spectral Representation and Scaling of Statistical Fluctuations

204| 205|Statistical fluctuations within $V_h = h^3$ from finite molecular sampling. Maxwell-Boltzmann with $\sigma_v^2 = k_B T/m$. $N_h \sim nh^3$, $n = N/V$. By i.i.d. propagation: 206|

$$207|\varepsilon_{\text{stat}}^2 \sim \frac{\sigma_v^2}{N_h} \sim \frac{k_B T/m}{nh^3} 208| \sim \frac{k_B T}{\rho} \cdot \frac{1}{h^3}, 209| \quad (15)$$

210|where $\rho = mn$ is mass density. Thus: 211|

$$212|\varepsilon_{\text{stat}} \sim C_{\text{stat}} \cdot N^{-1/2} h^{-3/2}, 213| \quad C_{\text{stat}} = \sqrt{\frac{k_B T}{\rho}}. 214| \quad (16)$$

215| 216|Note: $\varepsilon_{\text{stat}}$ is spatially white (δ -correlated), with flat wavenumber spectrum: 217|

$$218|E_{\text{stat}}(k) \sim \frac{\sigma_v^2}{N_h} \cdot \frac{k^2}{2\pi^2} \quad (\quad), 219| \quad (17)$$

220|The statistical contribution grows as k^2 (negligible at low k , potentially important at high k). 221| 222|

3.5 The Intersection Scale of Two Error Sources

223| 224|Define the intersection scale h_* where both errors are equal: 225|

$$226|\varepsilon_{\text{res}}(h_*) = \varepsilon_{\text{stat}}(h_*; N).227| \quad (18)$$

228| 229|From $\varepsilon_{\text{res}} \sim h^{1/3}$ (Kolmogorov) and $\varepsilon_{\text{stat}} \sim N^{-1/2}h^{-3/2}$ (thermal): 230|

$$231|h_*^{1/3} \sim N^{-1/2}h_*^{-3/2}232| \implies h_*^{11/6} \sim N^{-1/2}233| \implies h_* \sim N^{-3/11}.234| \quad (19)$$

235| 236|Conversely, for given h , a critical particle count $N_*(h) \sim h^{-11/3}$ exists: $N < N_* \rightarrow$ statistical dominates, $N > N_* \rightarrow$ truncation dominates. 237| 238|**Key observation:** h_* Kolmogorov $\eta = (\nu^3/\epsilon)^{1/4}$ $h < h_*$ $h > h_*$ “ ” —
 — $\varepsilon_{\text{res}}(h) \approx \varepsilon_{\text{stat}}(h)$ 239| 240|241|

4 Theorem 1: Resolution–Turbulence Unidentifiability Theorem (Strengthened)

242|243| 244|

4.1 Formalizing the Observation Operator

245| 246|

Definition 4.1 (Finite-Resolution Observation Operator). 247| 248|Let $h > 0$ be a resolution parameter. **Finite-Resolution Observation Operator** $\mathcal{O}_{\text{obs}h}$ is a mapping: 249|

$$250|\mathcal{O}_{\text{obs}h} : L^2(\mathbb{R}^3; \mathbb{R}^3) \rightarrow \mathbb{R}^{d_{\text{obs}}}, 251| \quad (20)$$

$$252| \quad d_{\text{obs}} \in \mathbb{N} \cup \{\infty\} \quad \mathcal{O}_{\text{obs}h} \quad \textbf{Resolution Constraint} \quad 253|$$

$$254|\|\mathcal{O}_{\text{obs}h}[\mathbf{u}] - \mathcal{O}_{\text{obs}h}[\mathbf{v}]\|_{\mathbb{R}^{d_{\text{obs}}}} 255| \leq L_{\mathcal{O}_{\text{obs}}} \cdot \|\mathcal{C}_h \mathbf{u} - \mathcal{C}_h \mathbf{v}\|_{L^2}, 256| \quad (21)$$

257| $L_{\mathcal{O}_{\text{obs}}} > 0$ Lipschitz $\mathcal{C}_h$ -h The Coarse-Graining Operator (??) h
 258|

259| 260|

4.2 Probabilistic Construction of the Three Worlds

261| 262|

Definition 4.2 (Probability Distributions of the Three Worlds). 263| 264|Let $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C$ be three physical worlds. Observations $\mathcal{O}_{\text{obs}h}[\mathbf{u}]$ are random vectors determined by each world’s velocity field and observation noise. Define: 265|

$$266|P_h^{\mathcal{W}_A}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}] + \boldsymbol{\xi} \in E), 267| \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), \quad (22)$$

$$268|P_h^{\mathcal{W}_B}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_N^{\mathcal{W}_B}] + \boldsymbol{\xi} \in E), 269| \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), \quad (23)$$

$$270|P_h^{\mathcal{W}_C}(E) := \mathbb{P}(\mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C}] + \boldsymbol{\xi} \in E), 271| \quad E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}}), 272| \quad (24)$$

273|where $\boldsymbol{\xi} \sim \mathcal{N}(0, \sigma_{\text{obs}}^2 \mathbf{I}_{d_{\text{obs}}})$ is independent measurement noise. Velocity fields: 274|

275|

- $\mathcal{W}_A \mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$ NS (??) 276|

• $\mathcal{W}_B$ NS $\mathbf{u}_N^{\mathcal{W}_B}$ N MD (??) (??) — “ ” **entirely** from truncation
 ε_{res} and fluctuation $\varepsilon_{\text{stat}}$. 277|

• $\mathcal{W}_C$ NS RANS $\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C} = \mathcal{G}_{\nu+\nu_t} * \mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$ ν_t $\mathbf{u}_{\text{RANS}}^{\mathcal{W}_C}$ $\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}$ 278|
 279|

280| 281|

4.3 Unidentifiability under Total Variation Distance

282| 283|

Theorem 4.3 (—). 284| 285| Let $P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}, P_h^{\mathcal{W}_C}$ be as in Def. ??, $\mathcal{O}_{\text{obsh}}$ satisfy Def. ??, with $\mathcal{W}_B$ parameters matched ($\varepsilon_{\text{res}}(h_B) \approx \varepsilon_{\text{res}}(h)$, $\varepsilon_{\text{stat}}(N) \approx \varepsilon_{\text{stat}}(h)$). Then: 286| 287|

288|

(i) **Total Variation Upper Bound.** $C_1, C_2 > 0$ $\alpha > 0$ Kolmogorov $\alpha = 1/3$
 $h > 0$ 289|

$$290| \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) 291| \leq C_1 \cdot h^\alpha + C_2 \cdot N^{-1/2}, 292| \quad (25)$$

$$293| \text{TV}(P, Q) := \sup_{E \in \mathcal{B}(\mathbb{R}^{d_{\text{obs}}})} |P(E) - Q(E)| \quad 294| 295|$$

(ii) **Limiting Identifiability.** 296|

$$297| \lim_{h \rightarrow 0, N \rightarrow \infty} \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) = 0. 298| \quad (26)$$

299| 300|

(iii) **Strict Unidentifiability at Finite Resolution.** $h > 0$ $N < \infty$ 301|

$$302| \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) > 0, 303| \quad (27)$$

304| and there exists an **unidentifiability interval** $\mathcal{I} = [h_{\min}, h_{\max}]$ (inertial range) such that for $h \in \mathcal{I}$: 305|

$$306| \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq \delta, 307| \quad (28)$$

308| where $\delta > 0$ is below any statistical test’s detection threshold. No observational test can distinguish these worlds. 309| 310|

(iv) **Three-World Unidentifiability.** Conclusion $(\mathcal{W}_A, \mathcal{W}_C)$ $(\mathcal{W}_B, \mathcal{W}_C)$ 311|

$$312| \max_{i,j \in \{A,B,C\}} \text{TV}(P_h^{\mathcal{W}_i}, P_h^{\mathcal{W}_j}) 313| \leq C \cdot (h^\alpha + N^{-1/2} + \delta_{\text{model}}), 314| \quad (29)$$

315| where δ_{model} is RANS model error (controllable via ν_t). 316|

317|

318| 319|

Proof. 320|[RIGOR: FULL] 321|(i) **TV** Lipschitz Pinsker 322|

$$323| \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{2\sigma} \|\mathcal{C}_h \mathbf{u}_{\text{NS}}^{\mathcal{W}_A} - \mathcal{C}_h \mathbf{u}_N^{\mathcal{W}_B}\|_{L^2}. 324|$$

325| Error Decomposition A, Lemma A.1-A.2 326|

$$327| \|\mathcal{C}_h \mathbf{u}_{\text{NS}}^{\mathcal{W}_A} - \mathcal{C}_h \mathbf{u}_N^{\mathcal{W}_B}\|_{L^2} \leq \varepsilon_{\text{res}}(h) + \varepsilon_{\text{stat}}(N, h) \leq C_{\text{res}} h^{1/3} + C_{\text{stat}} N^{-1/2} h^{-3/2}. 328|$$

329| $C_1 = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(2\sigma)$, $C_2 = L_{\mathcal{O}_{\text{obs}}} C_{\text{stat}} h_0^{-3/2}/(2\sigma)$ (??) 330| 331|(ii) **Limiting Identifiability.** $\lim_{h \rightarrow 0} \varepsilon_{\text{res}}(h) = 0$, $\lim_{N \rightarrow \infty} \varepsilon_{\text{stat}}(N, h) = 0$ $\text{TV} \rightarrow 0$ 332| 333|(iii) $h > 0, N < \infty$ $\text{TV} > 0$ $h \in [h_{\min}, h_{\max}]$ 334|

$$335|\text{TV} \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{2\sigma} \left(C_{\text{res}} h^{1/3} + C_{\text{stat}} N^{-1/2} h^{-3/2} \right) \leq \delta. 336|$$

337| D $P(D) \leq \frac{1}{2} + \frac{1}{2} \text{TV} \leq \frac{1}{2} + \delta/2$ $\delta < 1$ $\square$ 338| $\square$

339| 340|

Corollary 4.4 (). 341| 342|At any finite $h > 0$, no statistical test distinguishes true turbulence from truncation artifacts with probability $> 1/2 + Ch^{1/3}$, $C = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(4\sigma)$. 343|

344| 345| $\mathcal{O}_{\text{obs}h}$ Lipschitz ξ Gauss KL Gauss 346|

$$347|D_{\text{KL}}(P_h^{\mathcal{W}_A} \| P_h^{\mathcal{W}_B}) 348| \lesssim \frac{1}{\sigma_{\text{obs}}^2} \left\| \mathcal{O}_{\text{obs}h}[\mathbf{u}_{\text{NS}}^{\mathcal{W}_A}] - \mathcal{O}_{\text{obs}h}[\mathbf{u}_N^{\mathcal{W}_B}] \right\|^2 349| \lesssim \frac{L_{\mathcal{O}_{\text{obs}}}^2}{\sigma_{\text{obs}}^2} (\varepsilon_{\text{res}}(h)^2 + \varepsilon_{\text{stat}}(N)^2). 350| \quad (30)$$

351| 352| (??)-(??) $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ $C_1 = L_{\mathcal{O}_{\text{obs}}} C_{\text{res}}/(\sqrt{2}\sigma_{\text{obs}})$, $C_2 = L_{\mathcal{O}_{\text{obs}}} C_{\text{stat}}/(\sqrt{2}\sigma_{\text{obs}})$ $\alpha = 1/3$ (??) 353| 354|(ii) $\varepsilon_{\text{res}}(h) \rightarrow 0$ $h \rightarrow 0$ $\varepsilon_{\text{stat}}(N) \rightarrow 0$ $N \rightarrow \infty$ (??) $\mathbf{u}_{\text{true}}$ 355| 356|(iii) $\text{TV} > 0$ (h, N) $\varepsilon_{\text{res}}(h) > 0$ $\varepsilon_{\text{stat}}(N) > 0$ unidentifiability interval h h_* (??) TV 357| 358|(iv) 359|

$$360|\text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_C}) 361| \leq \text{TV}(P_h^{\mathcal{W}_A}, P_h^{\mathcal{W}_B}) + \text{TV}(P_h^{\mathcal{W}_B}, P_h^{\mathcal{W}_C}), 362| \quad (31)$$

363| $P_h^{\mathcal{W}_C}$ δ_{model} RANS (??) 364| 365| 366|

Corollary 4.5 (unidentifiability interval). 367| 368| Kolmogorov $\alpha = 1/3$ unidentifiability interval $\mathcal{I}$ $h_* \sim (\sigma_{\text{obs}}/L_{\mathcal{O}_{\text{obs}}})^{6/11} \cdot N^{-3/11}$ $O(h_*)$ TV 369|

$$370|\text{TV} \leq \frac{L_{\mathcal{O}_{\text{obs}}}}{\sqrt{2}\sigma_{\text{obs}}} 371| \sqrt{C_{\text{res}}^2 h_*^{2/3} + C_{\text{stat}}^2 N^{-1} h_*^{-3}} 372| = O(\max(h_*^{1/3}, N^{-1/2} h_*^{-3/2})). 373| \quad (32)$$

374|

375| 376| 377|

5 Theorem 2: Multi-Model Turbulence Audit Theorem

378| 379| 380|

5.1 Problem Setting

381| 382|In practice, multiple models predict the same flow; inter-model consensus indicates reliability. This theorem provides the probabilistic guarantee. 383| 384|

Definition 5.1 (Multi-Model Ensemble). 385| 386|Let $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$ be M distinct models, including: 387|

388| $\mathcal{M} = \{389|\text{RANS } k\text{-}\varepsilon, 390|\text{RANS } k\text{-}\omega \text{ SST}, 391|\text{LES Smagorinsky}, 392|\text{DES}, 393|\text{DNS } 394|\}$. 395| (33)

396| $\mathcal{M}_m$ $\mathbf{u}_m(t, \mathbf{x})$ $\mathbf{u}_{\text{DNS}}$ DNS $\Delta_{\text{DNS}} \lesssim \eta$ “ ” ground truth anchor 397|

398| 399|

Definition 5.2 (Systematic Bias). Model $\mathcal{M}_m$'s **systematic bias** for configuration Γ (in statistical steady state):

$$B_m(\Gamma) := \|\mathbb{E}[\mathbf{u}_m] - \mathbb{E}[\mathbf{u}_{\text{DNS}}]\|_{L^2(\Omega)}, \quad (34)$$

where $\Omega \subset \mathbb{R}^3$ is the flow domain, $\mathbb{E}[\cdot]$ denotes time average (steady) or ensemble average (unsteady).

5.2 Consensus Audit Theorem

Theorem 5.3 (Multi-Model Turbulence Audit Theorem). Let $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$ be M models. Assume:

(A1) **Bounded Bias:** there exists $L > 0$ such that $B_m(\Gamma) \in [0, L]$ for all Γ, m .

(A2) **Model Diversity (Weak Dependence):** $\{B_m\}_{m=1}^M$ α - $\alpha(k) \leq C\rho^k$
 $\rho \in (0, 1)$

(A3) **DNS Anchor Accessibility:** there exists $h_{\text{DNS}} \leq \eta$ such that $\mathbf{u}_{\text{DNS}}$ is a reliable truth approximation.

Then for any $\Delta > 0$:

(i) **Independent Model Bound (Strongest):** If model errors are *entirely* (different fundamental assumptions), then:

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\}\right) \leq \exp\left(-\frac{2M\Delta^2}{L^2}\right) \quad (35)$$

(ii) **Weak-Dependence Bound (General):** under α -mixing, there exists $C_\alpha > 0$ such that:

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\}\right) \leq C_\alpha \cdot \exp\left(-\frac{2M_{\text{eff}}\Delta^2}{L^2}\right), \quad (36)$$

where $M_{\text{eff}} = M/(1 + 2\sum_{k=1}^{M-1} \alpha(k))$ is the effective independent count.

(iii) **Consensus Reliability:** $\max_{i,j} \|\mathbf{u}_i - \mathbf{u}_j\|_{L^2} \leq \varepsilon$
 Δ

$$\mathbb{P}\left(\bigcap_{m=1}^M \{B_m > \Delta\} \mid \max_{i,j} \|\mathbf{u}_i - \mathbf{u}_j\|_{L^2} \leq \varepsilon\right) \leq \exp\left(-\frac{2M(\Delta - \varepsilon)^2}{L^2}\right), \quad (37)$$

valid when $\Delta > \varepsilon$.